

BMATS101 – MATHEMATICS-I (CSE)

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MODULE 5

LINEAR ALGEBRA

MATRICES, ROW TRANSFORMATIONS & RANK

1. Introduction to Linear Algebra

Linear Algebra is the study of **vectors, matrices, and systems of linear equations**.

It plays a central role in computer science and engineering, especially in:

- Data analysis and machine learning
- Computer graphics and transformations
- Network analysis
- Optimization problems
- Numerical methods

This module focuses on **matrix methods** for solving systems of equations and related concepts.

2. Matrices

2.1 Definition

A **matrix** is a rectangular array of numbers arranged in rows and columns.

If a matrix has m rows and n columns, it is called a matrix of order $m \times n$.

General form:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

3. Types of Matrices (Relevant to This Module)

- **Row matrix:** only one row
- **Column matrix:** only one column
- **Square matrix:** same number of rows and columns

- **Zero matrix:** all elements are zero
- **Identity matrix:** diagonal elements are 1, others 0

4. Elementary Row Transformations

Elementary row operations are used to simplify matrices and solve systems of equations.

4.1 Types of Row Transformations

1. Row interchange

$$R_i \leftrightarrow R_j$$

2. Row multiplication

$$R_i \rightarrow kR_i, k \neq 0$$

3. Row addition

$$R_i \rightarrow R_i + kR_j$$

These operations do **not change the solution** of a linear system.

5. Echelon Form of a Matrix

A matrix is in **row echelon form** if:

- All non-zero rows are above rows of all zeros
- Leading coefficient of each non-zero row is to the right of the one above it
- All entries below leading coefficients are zero

6. Reduced Row Echelon Form (RREF)

A matrix is in **reduced row echelon form** if:

- It is in row echelon form
- Leading entry of each non-zero row is 1
- All other entries in the leading column are zero

7. Rank of a Matrix

7.1 Definition

The **rank of a matrix** is defined as:

- The **maximum number of linearly independent rows or columns**, or
- The **number of non-zero rows** in its row echelon form.

$\text{Rank}(A) = \text{Number of non-zero rows in echelon form}$

8. Finding Rank Using Row Reduction

Example 1

Find the rank of:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$$

Solution

Apply row operations:

$$\begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \\ R_3 &\rightarrow R_3 - R_1 \\ \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix} \end{aligned}$$

Non-zero rows = 2

$$\boxed{\text{Rank}(A) = 2}$$

9. Systems of Linear Equations

A system of linear equations can be written in matrix form as:

$$AX = B$$

Where:

- A is the coefficient matrix
- X is the column matrix of variables
- B is the constant matrix

10. Augmented Matrix

The **augmented matrix** of the system $AX = B$ is written as:

$$[A \mid B]$$

Row operations are applied to the augmented matrix to study consistency and solutions.

11. Consistency of Linear Systems

A system of equations is:

- **Consistent** if at least one solution exists

- **Inconsistent** if no solution exists

The consistency depends on the ranks of matrices.

12. Rank Conditions for Consistency

Let:

- $\text{Rank}(A)$ = rank of coefficient matrix
- $\text{Rank}(A | B)$ = rank of augmented matrix

Then:

1. If

$$\text{Rank}(A) = \text{Rank}(A | B)$$

→ system is **consistent**

2. If

$$\text{Rank}(A) \neq \text{Rank}(A | B)$$

→ system is **inconsistent**

13. Types of Solutions (Consistent Systems)

If system is consistent:

- **Unique solution** if

$$\text{Rank}(A) = \text{number of variables}$$

- **Infinitely many solutions** if

$$\text{Rank}(A) < \text{number of variables}$$

GAUSS ELIMINATION & GAUSS-JORDAN METHODS

1. Introduction

Systems of linear equations can be solved efficiently using **matrix methods**.

Two important direct methods are:

- **Gauss Elimination Method**
- **Gauss-Jordan Method**

These methods use **elementary row transformations** to obtain the solution.

2. Gauss Elimination Method

2.1 Idea of the Method

Gauss elimination method reduces the given system of equations into an **upper triangular form**, from which the unknowns are obtained by **back substitution**.

3. Steps in Gauss Elimination Method

1. Write the system in augmented matrix form $[A | B]$.
2. Use elementary row transformations to convert the matrix into **row echelon form**.
3. Convert the matrix into an equivalent **upper triangular system**.
4. Solve the equations using back substitution.

4. Solved Example – Gauss Elimination Method

Example 1

Solve the system:

$$\begin{aligned}x + y + z &= 6 \\2x + 3y + 4z &= 20 \\3x + y + 2z &= 10\end{aligned}$$

Solution

Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 4 & 20 \\ 3 & 1 & 2 & 10 \end{array} \right]$$

Apply row operations:

$$\begin{aligned}R_2 &\rightarrow R_2 - 2R_1 \\R_3 &\rightarrow R_3 - 3R_1\end{aligned}$$
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & -2 & -1 & -8 \end{array} \right]$$

Next:

$$R_3 \rightarrow R_3 + 2R_2$$
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 3 & 8 \end{array} \right]$$

Back substitution:

$$3z = 8 \Rightarrow z = \frac{8}{3}$$
$$y + 2z = 8 \Rightarrow y = \frac{8}{3}$$

$$x + y + z = 6 \Rightarrow x = \frac{2}{3}$$

Final Solution

$$x = \frac{2}{3}, y = \frac{8}{3}, z = \frac{8}{3}$$

5. Gauss–Jordan Method

5.1 Idea of the Method

Gauss–Jordan method transforms the augmented matrix directly into **reduced row echelon form**, from which solutions are obtained **without back substitution**.

6. Steps in Gauss–Jordan Method

1. Write the augmented matrix $[A \mid B]$.
2. Apply row operations to make diagonal elements equal to 1.
3. Make all elements above and below the diagonal zero.
4. Read the solution directly.

7. Solved Example – Gauss–Jordan Method

Example 2

Solve the system:

$$\begin{aligned} 2x + y + z &= 10 \\ x + 3y + 2z &= 13 \\ 3x + y + 3z &= 14 \end{aligned}$$

Solution

Augmented matrix:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 10 \\ 1 & 3 & 2 & 13 \\ 3 & 1 & 3 & 14 \end{array} \right]$$

Row operations:

$$\begin{aligned} R_1 &\rightarrow \frac{1}{2}R_1 \\ \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 5 \\ 1 & 3 & 2 & 13 \\ 3 & 1 & 3 & 14 \end{array} \right] \\ R_2 &\rightarrow R_2 - R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 5 \\ 0 & \frac{5}{2} & \frac{3}{2} & 8 \\ 0 & -\frac{1}{2} & \frac{3}{2} & -1 \end{array} \right]$$

Continue to reduce:

$$R_2 \rightarrow \frac{2}{5}R_2$$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 5 \\ 0 & 1 & \frac{3}{5} & \frac{16}{5} \\ 0 & -\frac{1}{2} & \frac{3}{2} & -1 \end{array} \right]$$

Eliminate other entries to reach RREF.

Final solution:

$$x = 2, y = 3, z = 1$$

8. Comparison of Methods

Gauss Elimination:

- Uses back substitution
- Reaches upper triangular form

Gauss–Jordan:

- No back substitution
- Reaches reduced row echelon form

GAUSS–SEIDEL ITERATIVE METHOD (APPROXIMATE SOLUTION)

1. Introduction to Iterative Methods

In many engineering and computer science problems, the system of linear equations is too large or complex to solve using direct methods.

In such cases, **iterative methods** are used to obtain **approximate solutions**.

The most commonly used iterative method is the **Gauss–Seidel method**.

2. Gauss–Seidel Method

The Gauss–Seidel method is an iterative technique for solving a system of linear equations in which each variable is updated **immediately** after it is computed.

It is an improvement over simple iteration methods and converges faster in many cases.

3. Conditions for Convergence

The Gauss–Seidel method converges if:

- The coefficient matrix is **diagonally dominant**, or
- The system satisfies certain convergence criteria

4. Diagonally Dominant System

A system of equations is said to be **diagonally dominant** if, in each equation, the absolute value of the diagonal coefficient is greater than the sum of the absolute values of the other coefficients in that equation.

For example, in the equation:

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

Diagonal dominance requires:

$$|a_{11}| > |a_{12}| + |a_{13}|$$

5. Procedure of Gauss–Seidel Method

1. Rewrite each equation by solving for the diagonal variable.
2. Choose initial approximations for all unknowns.
3. Substitute the latest available values immediately.
4. Repeat the process until successive approximations are sufficiently close.

6. Mathematical Formulation

Consider the system:

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3\end{aligned}$$

Rewriting:

$$\begin{aligned}x_1 &= \frac{1}{a_{11}}(b_1 - a_{12}x_2 - a_{13}x_3) \\x_2 &= \frac{1}{a_{22}}(b_2 - a_{21}x_1 - a_{23}x_3) \\x_3 &= \frac{1}{a_{33}}(b_3 - a_{31}x_1 - a_{32}x_2)\end{aligned}$$

7. Solved Example – Gauss–Seidel Method

Example 1

Solve the system using Gauss–Seidel method:

$$\begin{aligned}10x + y + z &= 12 \\2x + 10y + z &= 13 \\2x + 2y + 10z &= 14\end{aligned}$$

Step 1: Rewrite the equations

$$\begin{aligned}x &= \frac{1}{10}(12 - y - z) \\y &= \frac{1}{10}(13 - 2x - z) \\z &= \frac{1}{10}(14 - 2x - 2y)\end{aligned}$$

Step 2: Initial approximations

Let:

$$x^{(0)} = 0, y^{(0)} = 0, z^{(0)} = 0$$

Step 3: First iteration

$$\begin{aligned}x^{(1)} &= \frac{1}{10}(12 - 0 - 0) = 1.2 \\y^{(1)} &= \frac{1}{10}(13 - 2(1.2) - 0) = 1.06 \\z^{(1)} &= \frac{1}{10}(14 - 2(1.2) - 2(1.06)) = 0.948\end{aligned}$$

Step 4: Second iteration

$$\begin{aligned}x^{(2)} &= \frac{1}{10}(12 - 1.06 - 0.948) = 0.9992 \\y^{(2)} &= \frac{1}{10}(13 - 2(0.9992) - 0.948) = 1.00536 \\z^{(2)} &= \frac{1}{10}(14 - 2(0.9992) - 2(1.00536)) = 0.999088\end{aligned}$$

Step 5: Approximate Solution

$$\boxed{x \approx 1, y \approx 1, z \approx 1}$$

8. Stopping Criterion

The iterations are stopped when:

$$|x^{(k)} - x^{(k-1)}| < \varepsilon$$

for all variables, where ε is a small positive number.

9. Advantages of Gauss–Seidel Method

- Faster convergence than simple iteration
- Requires less computational memory

- Suitable for large systems

10. Applications

- Network analysis
- Power systems
- Finite difference methods
- Numerical simulations

EIGENVALUES, EIGENVECTORS & RAYLEIGH'S POWER METHOD

1. Introduction to Eigenvalues and Eigenvectors

Eigenvalues and eigenvectors are fundamental concepts in linear algebra and are widely used in:

- Data analysis and machine learning
- Image processing
- Network analysis
- Stability analysis
- Markov processes

They help describe how a linear transformation acts along specific directions.

2. Eigenvalues and Eigenvectors

2.1 Definition

Let A be a square matrix of order n .

A non-zero vector X is called an **eigenvector** of A if:

$$AX = \lambda X$$

where λ is a scalar called the **eigenvalue** corresponding to the eigenvector X .

2.2 Interpretation

- Eigenvectors give **directions** that remain unchanged under the transformation A .
- Eigenvalues represent the **scaling factor** along those directions.

3. Characteristic Equation

Rewriting:

$$AX = \lambda X \Rightarrow (A - \lambda I)X = 0$$

For a non-trivial solution:

$$|A - \lambda I| = 0$$

This determinant equation is called the **characteristic equation**.

4. Finding Eigenvalues

Steps

1. Form the matrix $A - \lambda I$
2. Compute the determinant $|A - \lambda I|$
3. Solve the resulting polynomial equation for λ

Example 1

Find the eigenvalues of:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Solution:

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 \\ \lambda^2 - 4\lambda + 3 &= 0 \\ \lambda &= 1, 3 \end{aligned}$$

5. Finding Eigenvectors

For each eigenvalue λ , solve:

$$(A - \lambda I)X = 0$$

Example 2

Find eigenvectors for:

$$\begin{aligned} \lambda &= 3 \\ A - 3I &= \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \end{aligned}$$

Solving:

$$-x + y = 0 \Rightarrow y = x$$

Eigenvector:

$$X = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

6. Properties of Eigenvalues

- Sum of eigenvalues = trace of matrix
- Product of eigenvalues = determinant of matrix
- Eigenvalues of a triangular matrix are its diagonal elements

7. Rayleigh's Power Method

7.1 Purpose

Rayleigh's power method is an **iterative method** used to find:

- The **dominant eigenvalue** (largest in magnitude)
- Corresponding eigenvector

8. Principle of Power Method

Given a matrix A and an initial vector X_0 , repeated multiplication:

$$X_{k+1} = AX_k$$

causes the sequence X_k to converge to the eigenvector corresponding to the **largest eigenvalue**.

9. Steps of Rayleigh's Power Method

1. Choose an initial non-zero vector X_0 .
2. Multiply $X_1 = AX_0$.
3. Find the largest component of X_1 and normalize.
4. Repeat the process until convergence.
5. The ratio of successive vectors gives the dominant eigenvalue.

10. Solved Example – Power Method

Example 3

Find the dominant eigenvalue of:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Take initial vector:

$$X_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Iteration 1:

$$X_1 = AX_0 = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

Dominant value:

$$\lambda_1 = \frac{3}{1} = 3$$

Iteration 2:

$$X_2 = AX_1 = \begin{bmatrix} 9 \\ 9 \end{bmatrix}$$

Dominant value remains:

$$\lambda = 3$$

Thus dominant eigenvalue is:

$$\boxed{\lambda = 3}$$

11. Applications of Eigenvalues and Eigenvectors

- Google PageRank algorithm
- Facial recognition systems
- Image compression
- Stability analysis of systems
- Markov chain analysis